HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES
Actuarial Mathematics and Statistics

F78PA
Probability and Statistics A

Semester 1 - 2019/20
Duration: 2 hours

Question	Marks
1	12
2	15
3	9
4	13
5	11
6	16
7	9
Total Marks	85

Candidates must attempt ALL SEVEN questions.
To receive full credit you must show your work and explain your answers.

Approved electronic calculators may be used

AMS Formula Sheet and Cambridge Statistical Tables are provided

1. The cumulative distribution function of a random variable X is defined by the function

$$F_X(x) = \begin{cases} 0, & \text{if } x < 0, \\ 0.3 + 0.6x^2, & \text{if } 0 \leq x < 1, \\ 1, & \text{if } x \geq 1. \end{cases}$$

(a) Sketch the graph of $F_X(x)$ and state the properties that it must satisfy to be a valid cumulative distribution function. Comment on whether the random variable X is discrete, continuous or otherwise, giving reasons for your answer.

[5 marks]

(b) State ImX, the *image* of X.

[1 mark]

(c) Use the function $F_X(x)$ to calculate:

(i) $\mathbb{P}(X = 1)$;
(ii) $\mathbb{P}(0.5 < X < 1)$.

[2 marks]

(d) A certain type of pet insurance policy offered by a given company pays claims for veterinary bills made by a customer in any calendar year up to a maximum total of £1000 after which no further claims are paid.

Suppose that the company believes that the amount paid out (in multiples of £1000) to a randomly selected customer in any year can be described by the random variable X with cumulative distribution function $F_X(x)$ above.

Calculate $\mathbf{E}(X)$ and hence suggest the minimum annual premium that the company should charge for such a policy.

[4 marks]

[*Total* 12 marks]

2. Consider the following game in which a player pays $£0.50$ to roll 3 fair 6-sided dice, where the scores on the dice are mutually independent random variables.

- If 3 sixes are obtained the player wins a cash prize of $£50$.
- If 2 sixes are obtained then the player wins a cash prize of $£5$.
- Otherwise the player wins nothing.

(a) Let X be the random variable representing the prize money won by the player in a single play of the game. Calculate $\mathbf{E}(X)$, $\mathbf{E}(X^2)$ and $\mathrm{Var}(X)$.

[6 marks]

(b) Suppose that 100 players each play the game once and that outcomes are independent over the 100 players. Let $Y_i, i = 1, \ldots, 100$, denote the *profit* (*i.e.* prize money won minus the cost of playing the game) made by player i. Use the Central Limit Theorem to approximate the probability that $\sum_{i=1}^{100} Y_i$ exceeds $£50$.

[6 marks]

(c) Consider the following modification to the game.

- Players now pay $£c$ to play.
- If 3 sixes are obtained the player wins a cash prize of $£50$.
- If 2 sixes are obtained then the player wins a cash prize of $£5$.
- If 1 six is obtained the player has their payment of $£c$ refunded.
- Otherwise the player receives nothing.

Identify the value of c for which the player's expected profit is zero. [3 marks]

[$Total 15$ marks]

3. A computer scientist believes that the time T, measured in hours, required by a supercomputer to complete a certain kind of 'big data' computing job follows a Gamma$(5, 1)$ distribution. As the supercomputer's cooling system becomes less efficient (and hence uses energy more rapidly) as the job progresses she believes that the total energy, in kiloWatt-hours (kWhr) used by the cooling system is given by $Y = 3T^{3/2}$.

(a) Sketch a graph to illustrate the relationship between T and Y and state the image of Y. [2 marks]

(b) Find an expression for the probability density function of Y. [5 marks]

(c) For $y > 0$, express $\mathbb{P}(Y > y)$ in terms of F_T, the cumulative distribution function of T. [2 marks]

[$Total$ 9 marks]

[PLEASE TURN OVER]

4. Let X and Y be jointly continuous random variables such that the marginal density of X is given by

$$f_X(x) = \begin{cases} 0, & \text{if } x < 0, \\ 2e^{-2x}, & \text{if } x \geq 0, \end{cases}$$

and the conditional density of $Y|X$ is given by

$$f_{Y|X}(y|x) = \begin{cases} 0, & \text{if } y < 0, \\ xe^{-xy}, & \text{if } y \geq 0. \end{cases}$$

(a) Identify the marginal distribution of X and the conditional distribution of $Y|X$.

Give an expression for the joint probability density function of X and Y.

Comment on whether X and Y are independent, giving a reason for your answer.

[4 marks]

(b) Show that $\mathbb{P}(Y > y|X = x) = e^{-xy}$, $x, y > 0$ and, by applying the *Partition Theorem* or otherwise, show that, for $y > 0$,

$$\mathbb{P}(Y > y) = \frac{2}{(2+y)}$$

.

[5 marks]

Hint. You may assume that, for $\alpha, \beta > 0$,

$$\int_0^\infty x^{\alpha-1}e^{-\beta x}dx = \frac{\Gamma(\alpha)}{\beta^\alpha}$$

and that for integers n, $\Gamma(n) = (n-1)!$.

(c) Show that the conditional density $f_{X|Y}(x|y)$ is given by

$$f_{X|Y}(x|y) = \begin{cases} 0, & \text{if } x < 0, \\ (y+2)^2 xe^{-(y+2)x}, & \text{if } x \geq 0. \end{cases}$$

and identify this distribution.

[4 marks]

[*Total* 13 marks]

[PLEASE TURN OVER]

5. Suppose that in a certain tennis tournament, 7 players who have studied at Heriot-Watt University (HWU) have reached the final 16. Denote these 7 players by the numbers 1,..., 7 and the remaining (non-HWU) players by the numbers 8, ..., 15.

The identification of pairings for the subsequent 8 matches is to be done by first generating a random ordering of the numbers $\{1, 2, 3, \ldots, 16\}$ in such a way that all orderings are equally likely. If $(a_1, ..., a_{16})$ denotes the random ordering, then the 8 ties are formed from the pairings

$$\{(a_1, a_2), (a_3, a_4), \ldots, (a_{15}, a_{16})\}.$$

Let X be a random variable denoting the number of HWU players who are *not drawn* against another HWU player.

(a) Explain why the image of X is the set $\{1, 3, 5, 7\}$.

[2 marks]

(b) Let $I_i, i = 1, ..., 7$ denote indicator random variables such that $I_i = 1$ if and only if Player i is not drawn against another HWU player and $I_i = 0$ otherwise. Express X in terms of the variables $I_i, i = 1, \ldots, 7$.

[1 mark]

(c) Show that $\mathbf{E}(I_i) = \frac{8}{15}$, $\mathbf{E}(I_i^2) = \frac{8}{15}$, and $\mathbf{E}(I_i I_j) = \frac{56}{195}$ where $i = 1, \ldots 7, i \neq j$. [4 marks]

(d) Using the results of (b) and (c) above, calculate $\mathbf{E}(X)$ and $\mathbf{E}(X^2)$. [4 marks]

[*Total* 11 marks]

[PLEASE TURN OVER]

6. Suppose that $X_1, X_2, ..., X_n$ are independent, identically distributed random variables each distributed as Exp(λ) where $\lambda > 0$.

(a) Show that the moment generating function (m.g.f.) of X_1 is given by

$$M_{X_1}(\theta) = \frac{\lambda}{\lambda - \theta}.$$

[3 marks]

(b) Suppose that $Z = X_1 + X_2 + ... + X_n$. Using the result of (a) write down the moment generating function of the random variable Z and identify the distribution of Z.

[2 marks]

(c) (i) Given that $M_Y(\theta)$ is the m.g.f. of the random variable Y, define $K_Y(\theta)$, the cumulant generating function (c.g.f.) of Y. How you would you use $K_Y(\theta)$ to obtain the mean and variance of Y (assuming that both exist).

[2 marks]

(ii) Obtain the cumulant generating function of the random variable Z defined in (b) and obtain the mean and variance of Z.

[4 marks]

(d) State *Chebyshev's inequality* for a random variable X with $\mathrm{E}(X^2) < \infty$.

[2 marks]

(e) Suppose that $n = 10$ and $\lambda = 2$. By applying Chebyshev's inequality to the random variable $Z - 5$, where Z is the random variable in (b), find an upper bound for $\mathbb{P}(Z \geq 10)$.

[3 marks]

[*Total* 16 marks]

7. Let X and Y be random variables with means and μ_X, μ_Y respectively and variances σ_X^2, σ_Y^2 respectively.

(a) Define the covariance of X with Y, $\mathrm{Cov}(X, Y)$, and the correlation coefficient $\rho(X, Y)$. Describe the relationship between X and Y in the particular cases where $\rho(X, Y) = 1$ and $\rho(X, Y) = -1$ and identify the range of values that $\rho(X, Y)$ can take in general.

If $\mathrm{Cov}(X, Y) = 0$ can we deduce that X and Y are independent random variables?

[5 marks]

(b) Derive an expression for $\mathrm{Var}(X + Y)$ in terms of σ_X^2, σ_Y^2 and $\rho(X, Y)$. Discuss briefly how this variance is affected by the correlation coefficient $\rho(X, Y)$. [4 marks]

[*Total* 9 marks]

END OF PAPER